

scripts/check-api-app-sync.sh

The **CI gate** guaranteeing the on-device Lava API embed (liblavaapi.so, packaged by :core:apiengine into the :api-app APK) is built from EXACTLY the current lava-api-go source — no drift (§11.4.69 / §6.J).

What it does

1. Recomputes the live hash via scripts/compute-api-source-hash.sh.
2. Reads the committed manifest core/apiengine/src/main/resources/api-source.hash (the hash the last freshly-built .so was built from, written by build-cshared.sh).
3. If they differ, exits 1 loudly: on-device API embed is STALE vs lava-api-go – rebuild liblavaapi.so via build-cshared.sh, printing both hashes + the remediation.

Usage

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- Exit 0 + OK: ... when the manifest matches the live source.
- Exit 1 when stale (drift) or the manifest is missing.
- Exit 2 on a config error (hash script missing).

Remediation on failure

```
lava-api-go/scripts/build-cshared.sh    # rebuilds the .so AND  
                                       refreshes the manifest  
git add core/apiengine/src/main/resources/api-source.hash
```

Where it is wired

scripts/ci.sh runs it in BOTH --changed-only and --full modes (it is fast pure-bash + git, and a stale embed is a release blocker regardless of mode).

Falsifiability rehearsal (recorded)

Append a byte to a tracked embed-linked `.go` (e.g. `lava-api-go/internal/version/version.go`) → the gate exits 1 (live hash now differs from the manifest). Revert → exit 0. This proves the gate genuinely discriminates a drifted embed.

Constitutional notes

§6.R (no hardcoded values). §6.T.2 / §11.4.67 (bounded, read-only, no host mutation). No failure-swallow (§6.J) — a mismatch propagates non-zero. See `docs/ON_DEVICE_API.md` §5A and `docs/scripts/compute-api-source-hash.sh.md`.